Firewall Lab — pfSense 2.7.2

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1. Project Overview

This lab deploys pfSense 2.7.2-RELEASE as a full-featured enterprise firewall running on UTM (QEMU emulation) on macOS Apple Silicon. The firewall separates WAN and LAN zones, enforces traffic rules using named aliases, blocks insecure protocols, and implements automatic outbound NAT — replicating the same firewall logic used with Fortinet and Cisco firewalls in production environments.

Lab Environment

Firewall OS	pfSense 2.7.2-RELEASE (amd64) — FreeBSD 14.0
Hypervisor	UTM (QEMU emulation) on macOS Apple Silicon M1
CPU (emulated)	Intel Core Processor (Skylake) — 2 vCPUs
WAN Interface	em0 — 10.0.0.228/24 — DHCP — Gateway 10.0.0.1
LAN Interface	em1 — 10.0.0.50/24 — Static
Web GUI	http://10.0.0.50 — accessed from Mac browser
NAT Mode	Automatic outbound NAT (IPsec passthrough included)
Uptime	00 Hours 19 Minutes 53 Seconds at verification

2. Firewall Aliases

Aliases group IPs, networks, and ports into named objects — making firewall rules cleaner, more readable, and easier to maintain. This is how enterprise firewall engineers write production rules.

Alias Name	Type	Value	Description
LAN_NET	Network	10.0.0.0/24	LAN network range
ADMIN_HOST	Host	10.0.0.2	Admin workstation IP
BLOCKED_PORTS	Port	23	Telnet — insecure protocol
WEB_PORTS	Port	80, 443	HTTP and HTTPS traffic

3. Firewall Rules

LAN interface rules control what traffic is allowed in and out of the network. Rules are evaluated top to bottom — the first match wins. Block rules are placed before allow rules to ensure insecure protocols are always dropped.

Action	Protocol	Source	Port	Description
BLOCK	IPv4 TCP	Any	23 (Telnet)	Block insecure Telnet
PASS	IPv4 TCP	LAN_NET	80-443	Allow HTTP and HTTPS
PASS	IPv4 TCP/UDP	LAN_NET	53 (DNS)	Allow DNS resolution
PASS	IPv4 *	LAN subnets	Any	Default allow LAN to any

4. NAT Configuration

Automatic outbound NAT translates all LAN traffic to the WAN IP address when leaving the network. pfSense automatically created two rules — one for general traffic and one for ISAKMP (IPsec VPN passthrough). This mirrors real enterprise firewall NAT configurations.

NAT Mode	Automatic outbound NAT rule generation
Auto Rule 1	WAN — 10.0.0.0/24 → WAN address — port 500 (ISAKMP)
Auto Rule 2	WAN — 10.0.0.0/24 → WAN address — all ports
IPsec	Passthrough included automatically

5. Verification — pfSense Dashboard

pfSense 2.7.2-RELEASE dashboard showing system information — QEMU guest, Intel Skylake CPU (emulated), FreeBSD 14.0, AES-NI crypto, Kernel PTI enabled, and system on latest version confirmed.

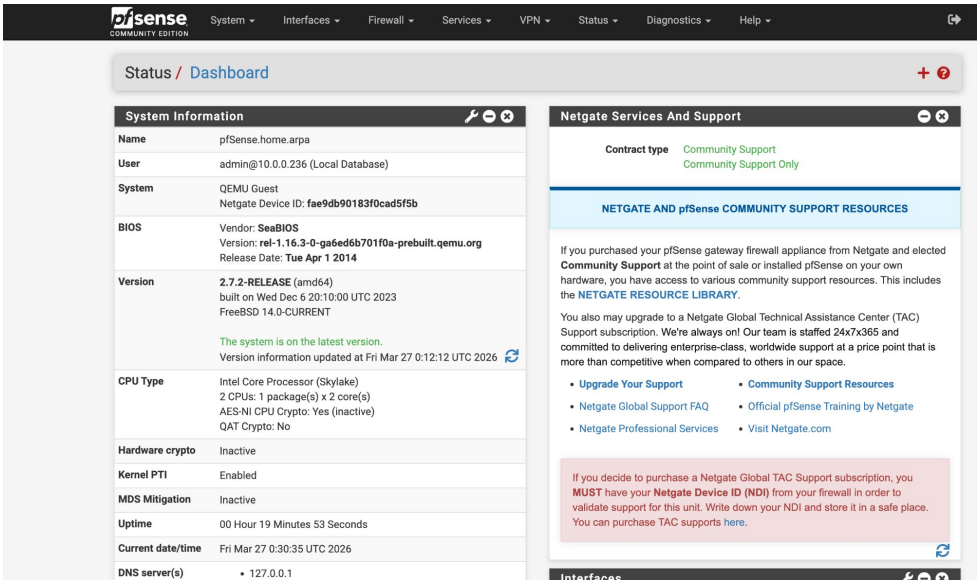

Figure 1: pfSense 2.7.2-RELEASE Dashboard — System Information

6. Verification — Firewall Aliases

Firewall Aliases page confirming all 3 aliases are created — ADMIN_HOST (Host: 10.0.0.2), BLOCKED_PORTS (Port: 23), and LAN_NET (Network: 10.0.0.0/24) — ready to be referenced in firewall rules.

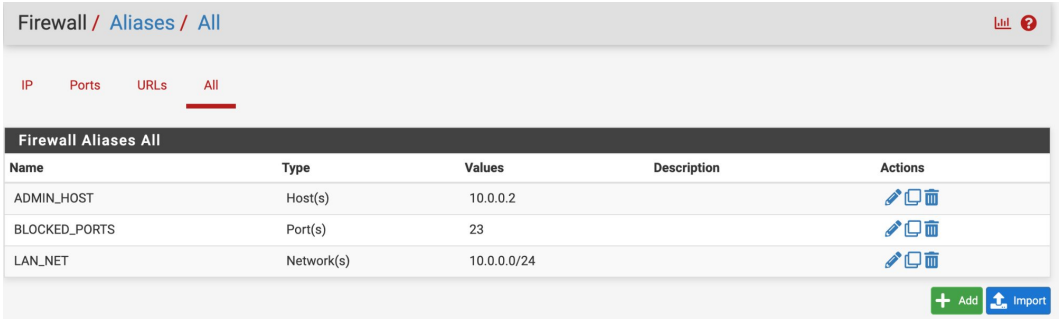

Figure 2: Firewall Aliases — ADMIN_HOST, BLOCKED_PORTS, LAN_NET configured

7. Verification — Firewall Rules

LAN firewall rules showing Block Telnet (port 23), Allow Web Traffic (80-443 using LAN_NET alias), Allow DNS (port 53 using LAN_NET alias), and default allow rules. Block rule marked with red X — confirming Telnet is actively blocked.

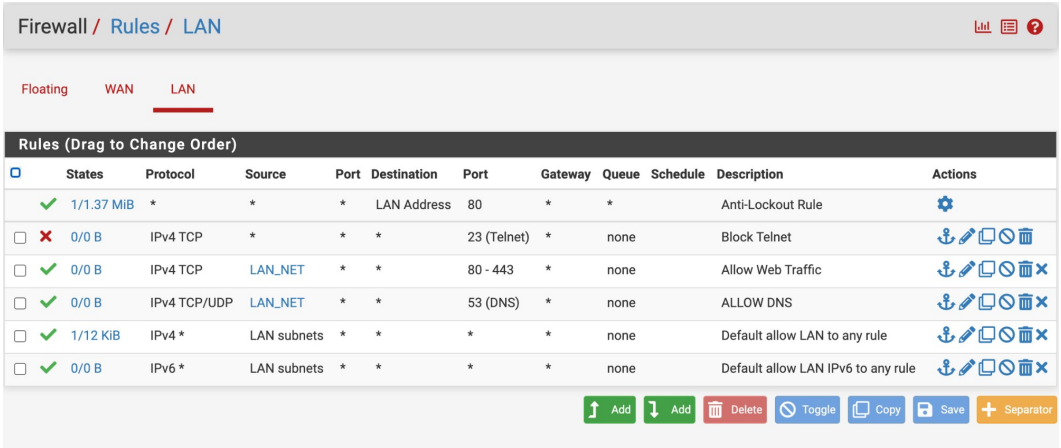

Figure 3: Firewall Rules LAN — Block Telnet, Allow Web Traffic, Allow DNS

8. Verification — NAT Outbound

NAT Outbound page showing Automatic outbound NAT mode selected with two auto-generated rules — ISAKMP passthrough and general outbound NAT for the 10.0.0.0/24 LAN subnet.

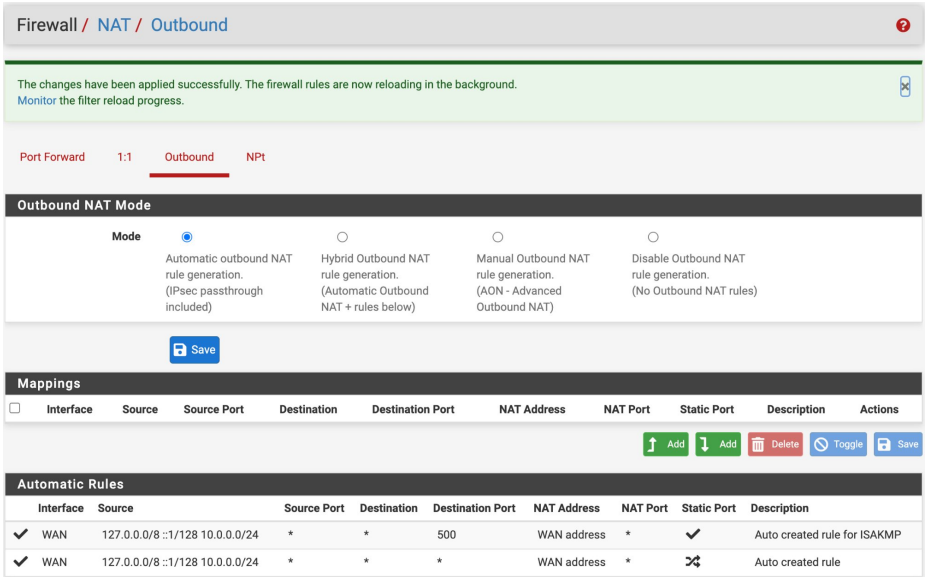

Figure 4: NAT Outbound — Automatic mode, ISAKMP passthrough, auto rules active

9. Verification — Interface Status

Interface status confirming both WAN (em0) and LAN (em1) are up with green status indicators. WAN showing 10.0.0.228 via DHCP with gateway 10.0.0.1. LAN showing static 10.0.0.50. Traffic counters confirm packets passing through.

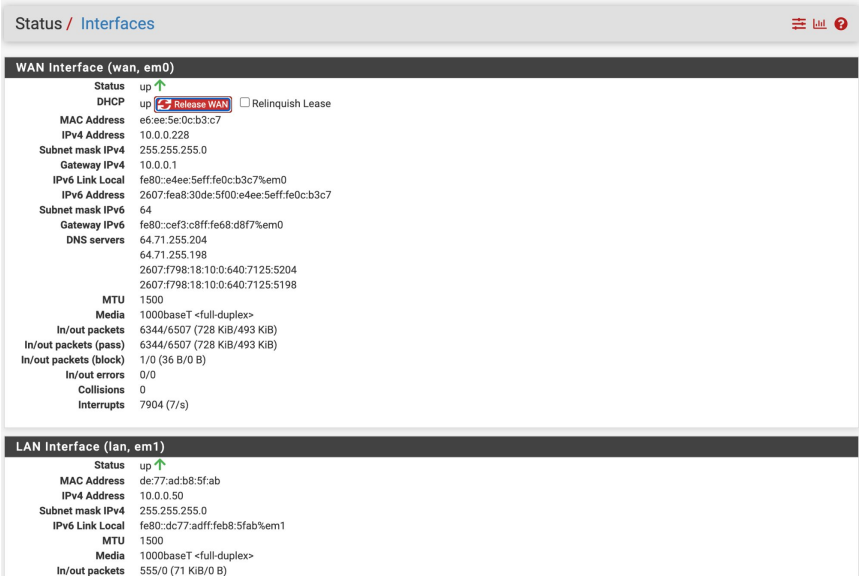

Figure 5: Interface Status — WAN 10.0.0.228 and LAN 10.0.0.50 both up

10. Troubleshooting Log

pfSense Download	pfsense.org no longer offers direct ISO downloads. Resolved using direct Netgate mirror: atxfiles.netgate.com/mirror/downloads/
ISO Extraction Error	macOS Archive Utility failed to extract .gz file. Resolved using Terminal: gunzip command extracted correctly.
UEFI Boot Loop	VM kept booting back into installer. Root cause: UEFI enabled in UTM conflicting with MBR partition. Fixed by unchecking UEFI Boot in UTM tweaks.
Disk Not Detected	ZFS installer showed 0 disks. Fixed by changing drive interface from VirtIO to IDE in UTM settings.
Web GUI Unreachable	LAN on Host-Only network not reachable from Mac browser. Fixed by changing LAN adapter to Bridged and setting static LAN IP to 10.0.0.50.
Package Manager Offline	Unable to retrieve package information for Snort. Root cause: WAN routing issue in UTM emulation. Documented as known limitation of QEMU bridged networking on Apple Silicon.

11. Skills Demonstrated

pfSense	`Installation, web GUI configuration, interface management`
Firewall Rules	`Stateful rules — block, pass, protocol filtering, port control`
Aliases	`Named objects for IPs, networks, ports — production rule style`

NAT	Automatic outbound NAT, IPsec passthrough, WAN translation
Network Zones	WAN/LAN separation, traffic segmentation, perimeter security
FreeBSD/QEMU	Emulation on Apple Silicon, BIOS/UEFI troubleshooting
Troubleshooting	6 real issues resolved — boot, networking, disk, GUI access

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